My Report

Chung-I LI

2024-03-04

Table of contents

Statistical Thinking

Reference: https://www.fharrell.com/post/rflow/

Summary Staistics

> **Note**
>
> Summary statistics provide a quick summary of data

```
library(Hmisc)
```

```
Warning: package 'Hmisc' was built under R version 4.3.2
```

```
Attaching package: 'Hmisc'
```

```
The following objects are masked from 'package:base':

    format.pval, units
```

```r
latex(describe(mtcars), file = "", caption.placement = "top")
```

mtcars
11 Variables   32 Observations

mpg

n	missing	distinct	Info	Mean	Gmd	.05	.10	.25	.50	.75	.90	.95
32	0	25	0.999	20.09	6.796	12.00	14.34	15.43	19.20	22.80	30.09	31.30

```
lowest : 10.4 13.3 14.3 14.7 15  , highest: 26   27.3 30.4 32.4 33.9
```

cyl

n	missing	distinct	Info	Mean	Gmd
32	0	3	0.866	6.188	1.948

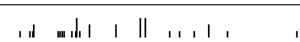
```
Value          4     6     8
Frequency     11     7    14
Proportion 0.344 0.219 0.438

For the frequency table, variable is rounded to the nearest 0
```

disp

n	missing	distinct	Info	Mean	Gmd	.05	.10	.25	.50	.75	.90	.95
32	0	27	0.999	230.7	142.5	77.35	80.61	120.83	196.30	326.00	396.00	449.00

```
lowest : 71.1 75.7 78.7 79   95.1, highest: 360  400  440  460  472
```

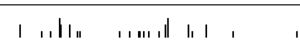
hp



n	missing	distinct	Info	Mean	Gmd	.05	.10	.25	.50	.75	.90	.95
32	0	22	0.997	146.7	77.04	63.65	66.00	96.50	123.00	180.00	243.50	253.55

```
lowest :  52  62  65  66  91, highest: 215 230 245 264 335
```

drat



n	missing	distinct	Info	Mean	Gmd	.05	.10	.25	.50	.75	.90	.95
32	0	22	0.997	3.597	0.6099	2.853	3.007	3.080	3.695	3.920	4.209	4.314

```
lowest : 2.76 2.93 3    3.07 3.08, highest: 4.08 4.11 4.22 4.43 4.93
```

wt

n	missing	distinct	Info	Mean	Gmd	.05	.10	.25	.50	.75	.90	.95
32	0	29	0.999	3.217	1.089	1.736	1.956	2.581	3.325	3.610	4.048	5.293

```
lowest : 1.513 1.615 1.835 1.935 2.14 , highest: 3.845 4.07  5.25  5.345 5.424
```

qsec

n	missing	distinct	Info	Mean	Gmd	.05	.10	.25	.50	.75	.90	.95
32	0	30	1	17.85	2.009	15.05	15.53	16.89	17.71	18.90	19.99	20.10

```
lowest : 14.5  14.6  15.41 15.5  15.84, highest: 19.9  20    20.01 20.22 22.9
```

---

vs

n	missing	distinct	Info	Sum	Mean	Gmd
32	0	2	0.739	14	0.4375	0.5081

---

am

n	missing	distinct	Info	Sum	Mean	Gmd
32	0	2	0.724	13	0.4062	0.498

---

gear

n	missing	distinct	Info	Mean	Gmd
32	0	3	0.841	3.688	0.7863

```
Value          3     4     5
Frequency     15    12     5
Proportion 0.469 0.375 0.156

For the frequency table, variable is rounded to the nearest 0
```

---

carb

n	missing	distinct	Info	Mean	Gmd
32	0	6	0.929	2.812	1.718

```
Value          1     2     3     4     6     8
Frequency      7    10     3    10     1     1
Proportion 0.219 0.312 0.094 0.312 0.031 0.031

For the frequency table, variable is rounded to the nearest 0
```

中文

Table 1

Objective: Epidemiologic and clinical research papers often describe the study sample in the first table. If well-executed, this "Table 1" can illuminate potential threats to internal and external validity. However, little guidance exists on best practices for designing a Table 1, especially for complex study designs and analyses. We aimed to summarize and extend the literature related to reporting descriptive statistics.

Reference: Hayes-Larson E, Kezios KL, Mooney SJ, Lovasi G. Who is in this study, anyway? Guidelines for a useful Table 1. J Clin Epidemiol. 2019 Oct;114:125-132. doi: 10.1016/j.jclinepi.2019.06.011. Epub 2019 Jun 20. PMID: 31229583; PMCID: PMC6773463.

In nearly every clinical trial or observational study, the researchers provide the baseline characteristics of the study participants in Table 1. This table will include any demographics the researchers recorded — age, geography, insurance status, income status, race/ethnicity, etc. — and any other characteristics that were relevant to the recruitment or randomization of the participants, such as comorbidities, severity of disease or prior exposure to a treatment.

reference: https://healthjournalism.org/glossary-terms/table-1/

Example 1:

```r
library(table1)
str(mtcars)
```

```
'data.frame':   32 obs. of  11 variables:
 $ mpg : num  21 21 22.8 21.4 18.7 18.1 14.3 24.4 22.8 19.2 ...
 $ cyl : num  6 6 4 6 8 6 8 4 4 6 ...
 $ disp: num  160 160 108 258 360 ...
 $ hp  : num  110 110 93 110 175 105 245 62 95 123 ...
 $ drat: num  3.9 3.9 3.85 3.08 3.15 2.76 3.21 3.69 3.92 3.92 ...
 $ wt  : num  2.62 2.88 2.32 3.21 3.44 ...
 $ qsec: num  16.5 17 18.6 19.4 17 ...
 $ vs  : num  0 0 1 1 0 1 0 1 1 1 ...
 $ am  : num  1 1 1 0 0 0 0 0 0 0 ...
 $ gear: num  4 4 4 3 3 3 3 4 4 4 ...
 $ carb: num  4 4 1 1 2 1 4 2 2 4 ...
```

```r
Mymtcars <- mtcars
label(Mymtcars$vs) <- "Engine"
Mymtcars$vs <- factor(Mymtcars$vs,
                      levels = c("0","1"),
                      labels = c("V-shaped","straight"))
label(Mymtcars$am) <- "Transmission"
Mymtcars$am <- factor(Mymtcars$am,
                        levels = c("0","1"),
                      labels = c("automatic","manual"))
str(Mymtcars)
```

```
'data.frame':   32 obs. of  11 variables:
 $ mpg : num  21 21 22.8 21.4 18.7 18.1 14.3 24.4 22.8 19.2 ...
 $ cyl : num  6 6 4 6 8 6 8 4 4 6 ...
 $ disp: num  160 160 108 258 360 ...
```

```
 $ hp  : num  110 110 93 110 175 105 245 62 95 123 ...
 $ drat: num  3.9 3.9 3.85 3.08 3.15 2.76 3.21 3.69 3.92 3.92 ...
 $ wt  : num  2.62 2.88 2.32 3.21 3.44 ...
 $ qsec: num  16.5 17 18.6 19.4 17 ...
 $ vs  : Factor w/ 2 levels "V-shaped","straight": 1 1 2 2 1 2 1 2 2 2 ...
 $ am  : Factor w/ 2 levels "automatic","manual": 2 2 2 1 1 1 1 1 1 1 ...
 $ gear: num  4 4 4 3 3 3 3 4 4 4 ...
 $ carb: num  4 4 1 1 2 1 4 2 2 4 ...
```

```r
table1(~ wt+hp+vs| am, data=Mymtcars)
```

	automatic	manual	Overall
	(N=19)	(N=13)	(N=32)
wt			
Mean (SD)	3.77 (0.777)	2.41 (0.617)	3.22 (0.978)
Median [Min, Max]	3.52 [2.47, 5.42]	2.32 [1.51, 3.57]	3.33 [1.51, 5.42]
hp			
Mean (SD)	160 (53.9)	127 (84.1)	147 (68.6)
Median [Min, Max]	175 [62.0, 245]	109 [52.0, 335]	123 [52.0, 335]
vs			
V-shaped	12 (63.2%)	6 (46.2%)	18 (56.3%)
straight	7 (36.8%)	7 (53.8%)	14 (43.8%)

```r
# help(table1)
```

Example 2:

```r
table1(~ wt+hp| vs*am, data=Mymtcars)
```

```
Get nicer `table1` LaTeX output by simply installing the `kableExtra` package
```

	automatic	manual	automatic	manual	automatic	manual
	(N=12)	(N=6)	(N=7)	(N=7)	(N=19)	(N=13)
wt						
Mean (SD)	4.10 (0.768)	2.86 (0.487)	3.19 (0.348)	2.03 (0.440)	3.77 (0.777)	2.41 (0.617)

	automatic	manual	automatic	manual	automatic	manual
Median [Min, Max]	3.81 [3.44, 5.42]	2.82 [2.14, 3.57]	3.22 [2.47, 3.46]	1.94 [1.51, 2.78]	3.52 [2.47, 5.42]	2.32 [1.51, 3.57]
hp						
Mean (SD)	194 (33.4)	181 (98.8)	102 (20.9)	80.6 (24.1)	160 (53.9)	127 (84.1)
Median [Min, Max]	180 [150, 245]	143 [91.0, 335]	105 [62.0, 123]	66.0 [52.0, 113]	175 [62.0, 245]	109 [52.0, 335]

```
table1(~ wt+hp| am*vs, data=Mymtcars)
```

```
Get nicer `table1` LaTeX output by simply installing the `kableExtra` package
```

	V-shaped	straight	V-shaped	straight	V-shaped	straight
	(N=12)	(N=7)	(N=6)	(N=7)	(N=18)	(N=14)
wt						
Mean (SD)	4.10 (0.768)	3.19 (0.348)	2.86 (0.487)	2.03 (0.440)	3.69 (0.904)	2.61 (0.715)
Median [Min, Max]	3.81 [3.44, 5.42]	3.22 [2.47, 3.46]	2.82 [2.14, 3.57]	1.94 [1.51, 2.78]	3.57 [2.14, 5.42]	2.62 [1.51, 3.46]
hp						
Mean (SD)	194 (33.4)	102 (20.9)	181 (98.8)	80.6 (24.1)	190 (60.3)	91.4 (24.4)
Median [Min, Max]	180 [150, 245]	105 [62.0, 123]	143 [91.0, 335]	66.0 [52.0, 113]	180 [91.0, 335]	96.0 [52.0, 123]

Missing Values

```
library(Hmisc)
library(DataExplorer)
```

```
Warning: package 'DataExplorer' was built under R version 4.3.2
```

```r
latex(describe(airquality), file = "", caption.placement = "top")
```

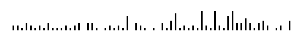
airquality
6 Variables   153 Observations

Ozone

n	missing	distinct	Info	Mean	Gmd	.05	.10	.25	.50	.75	.90	.95
116	37	67	0.999	42.13	35.28	7.75	11.00	18.00	31.50	63.25	87.00	108.50

```
lowest :   1    4    6    7    8, highest: 115 118 122 135 168
```

Solar.R



n	missing	distinct	Info	Mean	Gmd	.05	.10	.25	.50	.75	.90	.95
146	7	117	1	185.9	102.7	24.25	47.50	115.75	205.00	258.75	288.50	311.50

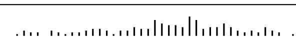
```
lowest :   7    8  13   14   19, highest: 320 322 323 332 334
```

Wind

n	missing	distinct	Info	Mean	Gmd	.05	.10	.25	.50	.75	.90	.95
153	0	31	0.997	9.958	3.964	4.60	5.82	7.40	9.70	11.50	14.90	15.50

```
lowest : 1.7  2.3  2.8  3.4  4   , highest: 16.1 16.6 18.4 20.1 20.7
```

Temp



n	missing	distinct	Info	Mean	Gmd	.05	.10	.25	.50	.75	.90	.95
153	0	40	0.999	77.88	10.74	60.2	64.2	72.0	79.0	85.0	90.0	92.0

```
lowest : 56 57 58 59 61, highest: 92 93 94 96 97
```

Month

n	missing	distinct	Info	Mean	Gmd
153	0	5	0.96	6.993	1.608

```
Value          5     6     7     8     9
Frequency     31    30    31    31    30
Proportion 0.203 0.196 0.203 0.203 0.196

For the frequency table, variable is rounded to the nearest 0
```

Day

n	missing	distinct	Info	Mean	Gmd	.05	.10	.25	.50	.75	.90	.95
153	0	31	0.999	15.8	10.26	2.0	4.0	8.0	16.0	23.0	28.0	29.4

```
lowest :  1  2  3  4  5, highest: 27 28 29 30 31
```

```
plot_missing(airquality)
```